

1. Given  $g(x, y) = 3e^{-2xy^4}$ , find  $\frac{\partial g}{\partial x}$ ,  $\frac{\partial^2 g}{\partial x^2}$ ,  $\frac{\partial^2 g}{\partial x \partial y}$ .
  
  
  
  
  
  
  
  
  
  
2. Given  $f(x, y) = \ln(x^2 - xy^3)$ , find  $\frac{\partial^2 f}{\partial x \partial y}$ .
  
  
  
  
  
  
  
  
  
  
3. A factory is producing 2 different products A and B. The monthly profit when  $x$  units of A and  $y$  units of B are produced per month is given by the function  $f(x, y) = x^{\frac{2}{3}}y^{\frac{1}{3}}$ . Currently the factory is producing 8 units of A and 27 units of B per month.
  - (a) Find the current profit, namely  $f(8, 27)$ .
  - (b) Find the current marginal profit with respect to A and B, namely  $\frac{\partial f}{\partial x}(8, 27)$  and  $\frac{\partial f}{\partial y}(8, 27)$ .
  - (c) Based on part (b), if the factory is going to produce one more unit of product, should the factory produce more product A or B to get larger profit? Estimate the new monthly profit using linear approximation.